

Section A: Pure Mathematics [40 marks]

- 1 Without using a calculator, solve the inequality $\frac{x^2 - 2x + 1}{-2 + 5x - 2x^2} \leq 0$. [3]

Hence solve $\frac{x^2 - 2|x| + 1}{-2 + 5|x| - 2x^2} \leq 0$. [2]

- 2 The complex numbers z and w satisfy the simultaneous equations

$$iw + z = 1 \quad \text{and} \quad zw^* = 2 - 6i.$$

Find z and w . [5]

- 3 Functions f and g are defined by

$$f : x \mapsto 2 - \frac{1}{(x-1)(x-2)}, \quad x \in \mathbb{R}, x \neq 1, x \neq 2,$$

$$g : x \mapsto \sin x, \quad x \in \mathbb{R}, -\frac{\pi}{2} < x < \frac{\pi}{2}.$$

- (i) Explain why the composite function fg exists. Find $fg(x)$ and state the domain and range of fg . [4]
 - (ii) Sketch the graph of f and explain why the function f^{-1} does not exist. [2]
 - (iii) The domain of f is now further restricted to $(-\infty, a)$. Find the largest value of a such that the function f^{-1} exists. [1]
 - (iv) Using your answer to part (iii), define f^{-1} in a similar form. [3]
- 4 Relative to the origin O , the position vectors of points A , B and N are $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{n}$ respectively. It is given that N lies on AB , between A and B , such that $\mathbf{a} \cdot \mathbf{gn} = \mathbf{b} \cdot \mathbf{gn}$ and $\angle AON$ and $\angle BON$ are 30° and 60° respectively.
- (i) Show that AB is perpendicular to ON . [2]
 - (ii) State the geometrical meaning of $|\mathbf{a} \times \mathbf{n}|$. [1]
 - (iii) State the geometrical meaning of $\left| \mathbf{a} \times \frac{\mathbf{n}}{|\mathbf{n}|} \right|$ and show that its value is $k|\mathbf{n}|$, where k is an exact value to be found. [2]
 - (iv) Find the ratio $AN : NB$ and express $\mathbf{n}$ in terms of $\mathbf{a}$ and $\mathbf{b}$. [3]

- 5 (a) The displacement of a particle, t seconds after leaving a fixed point O , is modelled by the curve C with parametric equations $x = e^t - 1, y = t - e^t$.
- (i) The normal to the curve at the point $P(e^p - 1, p - e^p)$ meets the y -axis at the point Q . Find the coordinates of Q . [4]
- (ii) Find the rate of change of the particle's y -coordinate when $y = -5$. [2]
- (b) The region R is bounded by the curve with equation $y = \frac{x}{1+x^2}$, the line $x = \sqrt{3}$ and the x -axis. Using the substitution $x = \tan \theta$, find the exact volume generated when R is rotated completely about the x -axis. [6]

Section B: Probability and Statistics [60 marks]

- 6 A bag contains 3 red balls and 2 white balls, which are indistinguishable except for their colours. Two balls are drawn at random and without replacement from the bag. If the balls are of different colours, two fair coins are tossed and the number of heads is recorded; if the balls are of the same colour, a fair coin is tossed and the number of heads is recorded. The random variable X is the number of heads recorded.
- (i) Show that $P(X = 0) = \frac{7}{20}$. [2]
- (ii) Find the probability distribution of X . [2]
- (iii) Find $E(X)$ and $\text{Var}(X)$. [2]
- (iv) Find the probability that the balls are of different colours given that no heads are obtained. [2]
- (v) Find the probability that either no heads are obtained or the balls obtained are of the same colour or both. [2]
- 7 Ten cards each has a single letter or a digit written on them. The letters are A, B, C, D, E and the digits are $1, 2, 3, 4, 5$.
- (i) Find the number of ways in which the 10 cards can be divided into two sets with each set consisting of 5 cards. [1]
- Find the number of ways of arranging the ten cards in a row such that
- (ii) all the consonants are next to each other and all the digits are next to each other, [2]
- (iii) the letters must be in alphabetical order from left to right. [2]
- The ten cards are now arranged at random in a circle.
- (iv) Find the probability that no two odd digits are next to each other. [3]

- 8 A manufacturer produces a large number of coloured glowing sticks, which are packed in packets of 8. Each packet consists of randomly chosen coloured glowing sticks. On average, 12% of the glowing sticks are green.
- (i) Explain the significance of 'large number' in the first sentence. [1]
 - (ii) Find the probability that a randomly chosen packet of glowing sticks contains at least 3 green glowing sticks. [2]
 - (iii) Ian buys 40 randomly chosen packets of glowing sticks for a concert. Find the probability that fewer than 6 of these packets contain at least 3 green glowing sticks. [2]

On average, the proportion of glowing sticks that are blue is p . It is known that the modal number of blue glowing sticks in a packet is 3.

- (iv) Use this information to find exactly the range of values that p can take. [4]
- 9 A new Internet service provider in the market, Y-Fai, decides to investigate the effect of the distance from its router on its wifi signal. The most convenient way to express wifi signal strength is by using dBm, which stands for decibels relative to a milliwatt. The signal strength measured in dBm is negative in value and a value closer to 0 signifies a stronger wifi signal. An employee measures the signal strength (y dBm) at various fixed distances away from its router (x m) as follows.

x	1	1.5	2	2.5	3	10	15
y	-30	-60	-100	-76	-79	-83	-88

- (i) Draw a scatter diagram for these values. On your diagram, circle the data point that seems to be unexpected and suggest a possible reason for the anomaly. [2]

For parts (ii) and (iii) of this question, you should **exclude** the anomaly.

- (ii) Explain from your scatter diagram why the relationship between x and y should not be modelled by an equation of the form $y = ax + b$. [1]
- (iii) Which of the formulae $y = \frac{c}{x} + d$ and $y = \frac{e}{x^2} + f$, where c , d , e and f are constants, is the better model for the relationship between y and x ? Explain fully how you decided, and find the constants for the better formula. [4]
- (iv) Use the formula you chose from part (iii) to estimate the signal strength when the distance away from its router is 5 m. Explain why you would expect this estimate to be reliable. [2]

- 10** At an early stage in analysing the marks scored by a large number of candidates in an examination paper, the Examination Board claims that the mean mark of candidates is 49. To test this claim, a random sample of 100 candidates is taken and the marks, x , are summarised as follows:

$$\sum(x-49) = 313 \text{ and } \sum(x-49)^2 = 23\,280.$$

- (i) Suggest a reason why, in this context, the given data is summarised in terms of $(x-49)$ rather than x . [1]
- (ii) Calculate unbiased estimates of the population mean and variance. [2]
- (iii) Test, at the 5% significance level, whether the Examination Board's claim is valid. [4]
- (iv) State, giving a valid reason, whether the Examination Board needs to make any assumptions about the population in order for the test to be valid. [1]

It is subsequently found that the population standard deviation for the paper is 11. The Examination Board now decides to test the claim that the mean mark of the candidates has increased. A new random sample of 50 candidates is chosen and the mean is k . It is assumed that the marks of the candidates are normally distributed. A test at the 4% significance level indicates that the mean mark of candidates has not increased.

- (v) Find the set of possible values of k . [4]

- 11** The lifespan of light bulbs manufactured by OceanDoor is normally distributed with a mean of 800 hours and a standard deviation of 70 hours. The lifespan of light bulbs manufactured by DaRice is also normally distributed with a standard deviation of 80 hours. It is known that there is a 50% chance that the lifespan of DaRice light bulbs is less than 700 hours.

- (i) A random sample of six DaRice light bulbs is chosen for quality check. Find the probability that the mean lifespan of the six DaRice light bulbs differs from half the lifespan of a OceanDoor light bulb by at least 200 hours. [4]
- (ii) Find the least value of n such that the probability that the average lifespan of n OceanDoor light bulbs is at most 825 hours is at least 0.99. [2]
- (iii) Find the probability that the total lifespan of a randomly selected OceanDoor light bulb and a randomly selected DaRice light bulb is more than 1450 hours. [2]
- (iv) Find the probability that a randomly selected OceanDoor light bulb and a randomly selected DaRice light bulb each has lifespan more than 725 hours. [2]
- (v) Explain why the value in (iii) is larger than (iv). [1]
- (vi) State an assumption needed for all your above calculations to be valid [1]

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